

BOYUE ZHANG

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EDUCATION

- **Zhejiang University** Sep. 2023 - Jun. 2027 (Expected)
B.S. in Biological Sciences, Bioinformatics track, GPA: 3.70/4.0 Hangzhou, China
 - **Coursework:** Computational Biology (98/100), Biophysics (96/100), Bioinformatics (97/100)
 - **Honors:** Professional Excellence Scholarship, Academic Excellence Model
- **The Chinese University of Hong Kong** Sep 2026 - Dec 2026 (Upcoming)
Exchange Student, International Asian Studies Programme (IASP) Hong Kong, China

PROJECTS

- **Walter and Eliza Hall Institute (WEHI) InSPIRE Research Intern** Jun 2026 - Present
Tools: Flow Cytometry, Degranulation Assay, HEK293T Culture, GROMACS, ΔG Prediction, Rosetta, ESM2 [🌐]
 - Supervised by Dr. Emma Petley; PIs: Prof. Matt Call & A/Prof. Melissa Call, Fully-funded InSPIRE Program
 - **Design:** Investigated whether redesigning a CAR-T costimulatory receptor (4-1BB ζ) TMD into its natural trimeric form improves T cell activation, based on the lab's proCAR3 design hypothesis
 - **Wet Lab (Functional Validation):** Collaborated to optimize 4 TMD variants and executed flow cytometry and degranulation assays against 2 tumor cell lines, confirming surface expression (73–74%) and T cell activation.
 - **Dry Lab (Computational Screening):** Designed and executed a four-route, 67 μ s simulation study (all-atom MD, umbrella sampling/SMD, MARTINI 3 coarse-grained self-assembly, insertion PMF) to separate assembly from membrane insertion as causes of the observed expression differences; utilized Rosetta/ESM2 to screen mutations for improved expression
- **Computational Design of CrtW and CrtZ for Enhanced Astaxanthin Biosynthesis** Sep 2025 - Present
Tools: Rosetta, GROMACS, ESM2, PyMOL, AlphaFold3, HPLC [🌐]
 - Advised by Dr. Yangwei Jiang, Qiushi Chair Prof. Ruhong Zhou's Lab, School of Life Sciences, Zhejiang University
 - **Design (268 candidates):** Engineered CrtW (β -carotene ketolase), a rate-limiting enzyme in astaxanthin biosynthesis, combining structure-based (Rosetta FuncLib) and sequence-based (ESM2, ThermoMPNN) design to screen 320 residues
 - **Simulation (8 μ s MD):** Evaluated the top 6 candidates; G142P+I192L formed the most extended substrate positioning and most stable ligand-binding pocket, outperforming static structural predictions with no loss of protein stability
 - **Validation (+46.7% yield):** Took the top mutant into the wet lab, confirming the predicted improvement in vitro over wild-type
- **LumaManta (ZJU-China iGEM 2025): Technical Lead** Aug 2024 - Oct 2025
Tools: CRISPR-i/a, Optogenetics, Confocal Microscopy, Python, R, GROMACS, AlphaFold3 [🌐] [🔗]
 - Advised by Prof. Ming Chen and Assoc. Prof. Fan Yang, School of Life Sciences, Zhejiang University
 - **Leadership:** Led a 14-member interdisciplinary team (wet lab, dry lab, hardware, software) to a Silver Medal, building an all-in-one biosensing device for real-time water pollutant and eDNA monitoring
 - **Biosensor design:** Engineered *BiChromaLogic*, a light-controlled genetic circuit that converts 2 light inputs into 4 distinct sensor proteins on demand, using CRISPR-based logic gates (169% activation, 92% repression efficiency)
- **Sex Dimorphism in DEN/TCPOBOP-Induced Hepatocellular Carcinoma** Mar 2026 - Present
Tools: R (DESeq2, WGCNA), Python, Transcriptomics, Metabolomics
 - Advised by Dr. Wenbin Zhou, Prof. Meihua Sui's Lab, School of Medicine, Zhejiang University
 - **Multi-omics analysis:** Analyzed transcriptomic and metabolomic data across hepatitis, cirrhosis, and HCC stages from 174 mice, identifying 12 core sex-biased metabolites in bile acid, tryptophan, and acylcarnitine pathways
 - **Hormone rescue (+97% TUDCA):** Found estrogen supplementation reversed the bile acid marker TUDCA and rescued half of the male-female gap in a composite disease score

- **Network integration (155 links):** Linked WGCNA/PPI hub genes to metabolomic findings, revealing candidate drivers of sex-biased liver disease progression
- **Provincial Research Grant (SRTP): A Sustainable Pigment Biosynthesis System** March 2025 - Feb 2026
Tools: GROMACS, LC-MS, SnapGene, GSMM, Python, R, Optogenetics 
 ◦ *Advised by Prof. Ming Chen, School of Life Sciences, Zhejiang University*
 - **Enzyme & metabolic engineering:** Identified a vioE mutant (+32% activity) via semi-rational design and MD simulation, and two gene knockouts (+48%/+23% yield) via genome-scale metabolic modeling
 - **Genetic circuit design:** Built dual orthogonal quorum-sensing circuits and a blue-light-inducible split-Cre system to enable programmable, reversible switching of pigment output

HONORS AND AWARDS

- **WEHI InSPIRE Scholarship** 2026
Walter and Eliza Hall Institute of Medical Research — 8,000 AUD; competitively selected, ZJU-nominated
- **iGEM 2025 Silver Medal** Oct 2025
International Genetically Engineered Machine Competition — Technical Lead, ZJU-China LumaManta 
- **Grand Prize, National Synthetic Biology Innovation Competition** Aug 2025
Chinese Society for Biotechnology & Shenzhen Institute of Synthetic Biology — 1st place nationally
- **Gold Medal, 18th “Dandelion” Undergraduate Innovation Competition** Mar 2026
Zhejiang University — CTO, LumaManta eDNA-based aquatic pathogen detection
- **First Prize (Champion), 2nd “Forbel Cup” National Pet Industry Innovation Competition** Jan 2025
Shanghai Society of Animal Husbandry & Veterinary Medicine — Led “Pet Smart Health” team
- **Provincial SRTP Research Grant** 2025
Zhejiang Provincial Department of Science and Technology — 12,000 RMB
- **5th Place, Stand-Up Paddleboard 200m, University Sports Championship** Jun 2025
Zhejiang University
- **Academic Honors & Scholarships** 2023–2025
Zhejiang University
 - Professional Excellence Scholarship (2023–2024)
 - Third-Class Scholarship (2024–2025)
 - Innovation & Entrepreneurship Model; Academic Excellence Model (2024–2025)

SKILLS

- **Programming Languages:** Python, R, Bash/Shell, C
- **Computational Biology & Bioinformatics:** Rosetta, AlphaFold3, GROMACS, ESM2, ThermoMPNN, PyMOL, SnapGene, CHARMM-GUI, MEGA
- **Data science & omics:** Transcriptomics, metabolomics, WGCNA, DESeq2, TCGA, GEO, scikit-learn, pandas, NumPy, SciPy
- **Molecular & Synthetic Biology:** CRISPR-i/a, Optogenetics, Gibson Assembly, Molecular Cloning, Site-Directed Mutagenesis, GSMM
- **Cell Biology & Protein:** Human Primary T Cell / Jurkat / HEK293T Culture, Viral Production & Purification, Flow Cytometry, Degranulation Assay, Cell Counting, Protein Expression & Purification, Western Blot, Immunofluorescence, Confocal Microscopy
- **Languages:** English (IELTS 6.5), Chinese (Native)

LEADERSHIP AND VOLUNTEER EXPERIENCE

- **Class Monitor** Sep 2023 - Jul 2025
School of Life Sciences & College of Yunfeng, Zhejiang University 
 ◦ Led 30+ students across two cohorts, organizing 10+ career seminars and coordinating peer mentoring for 15+ students
 ◦ Managed volunteer programs including animal welfare, West Lake conservation, and Hangzhou Zoo docent training